

ARCEN, Netherlands

WMO: 063910

Lat: 51.497N

Lon: 6.196E

Elev: 19

StdP: 101.10

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.7	-4.6	-10.4	1.6	-4.5	-7.7	2.0	-3.5	10.6	9.1	9.5	9.1	2.4	70	0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	30.3	20.1	27.9	19.4	26.0	18.4	21.3	27.5	20.4	26.0	19.4	24.3	3.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.1	23.5	18.4	13.3	22.3	17.6	12.7	21.4	61.9	27.7	58.6	26.3	55.4	24.3	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.4	6.5	DB	-9.2	34.5	3.2	2.3	-11.5	36.2	-13.3	37.6	-15.1	38.9	-17.4	40.6	(4)
(5)			WB	-9.6	23.4	3.1	1.0	-11.8	24.1	-13.6	24.7	-15.3	25.2	-17.6	26.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	3.1	4.0	6.6	10.3	14.1	17.0	18.8	18.2	15.4	11.2	7.1	4.0	(6)	
	DBStd	6.61	4.22	3.93	3.62	3.79	3.57	3.26	3.28	2.86	2.92	3.51	3.50	4.18	(7)	
	HDD10.0	861	214	169	114	42	6	0	0	0	1	28	99	189	(8)	
	HDD18.3	2848	471	401	363	241	138	64	33	37	98	220	338	444	(9)	
	CDD10.0	1174	1	2	10	52	133	209	274	253	161	66	11	3	(10)	
	CDD18.3	120	0	0	0	1	7	23	48	32	9	1	0	0	(11)	
	CDH23.3	1153	0	0	0	19	87	237	457	283	67	3	0	0	(12)	
	CDH26.7	338	0	0	0	2	14	65	153	91	13	0	0	0	(13)	
(14) Wind	WSAvg	3.0	3.7	3.6	3.3	3.0	3.0	2.7	2.7	2.5	2.5	3.0	3.1	3.6	(14)	
(15) Precipitation	PrecAvg	772	65	60	51	44	59	70	75	78	61	65	70	74	(15)	
	PrecMax	1050	148	132	102	109	120	230	155	162	158	177	114	163	(16)	
	PrecMin	566	5	10	4	0	13	13	23	24	11	28	8	30	(17)	
	PrecStd	98	31	30	28	23	27	42	35	43	40	29	27	31	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.3	14.6	19.6	26.2	28.6	31.8	34.1	33.0	28.6	22.8	16.9	13.7	(19)	
		MCWB	11.0	9.9	12.1	15.2	19.0	20.6	20.8	20.9	19.4	17.2	14.2	11.6	(20)	
	2%	DB	11.6	12.2	16.6	22.6	25.6	28.5	30.5	29.2	25.4	19.5	14.7	12.5	(21)	
		MCWB	10.1	9.4	11.0	14.2	16.7	19.3	20.0	20.3	18.7	15.9	12.6	10.8	(22)	
	5%	DB	10.3	10.9	14.3	19.7	23.6	26.0	27.9	26.4	22.9	17.6	13.3	11.2	(23)	
		MCWB	8.8	8.7	10.2	13.0	15.8	18.4	19.5	19.1	17.7	14.8	11.6	9.8	(24)	
	10%	DB	8.8	9.5	12.2	17.2	21.4	23.8	25.8	24.1	20.7	16.1	12.1	9.8	(25)	
		MCWB	7.5	7.8	9.1	11.9	15.0	17.4	18.8	18.1	16.4	13.9	10.6	8.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.1	13.1	16.3	20.1	22.5	22.7	22.7	20.6	17.7	14.3	12.2	(27)	
		MCDB	12.5	12.9	17.4	22.9	26.3	29.3	29.9	30.2	25.9	21.3	15.9	13.1	(28)	
	2%	WB	10.2	10.0	11.8	14.8	18.3	20.6	21.3	21.0	19.4	16.2	13.0	11.1	(29)	
		MCDB	11.4	11.6	15.1	20.9	23.4	26.3	27.6	27.0	24.0	18.9	14.5	12.3	(30)	
	5%	WB	8.9	9.0	10.6	13.6	16.8	19.3	20.4	19.9	18.2	15.3	11.9	9.9	(31)	
		MCDB	10.2	10.6	13.6	18.5	21.5	24.2	26.3	25.0	22.1	17.3	13.2	11.0	(32)	
	10%	WB	7.6	7.8	9.6	12.4	15.6	18.2	19.4	18.9	17.1	14.2	10.6	8.6	(33)	
		MCDB	8.7	9.5	12.0	16.6	20.0	22.4	24.3	22.9	20.0	15.8	11.9	9.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.8	6.0	8.1	10.1	10.2	10.4	10.0	9.9	9.3	7.2	5.5	4.5	(35)	
		MCDBR	5.3	7.4	11.9	14.3	13.9	14.3	14.3	14.1	12.1	8.7	6.4	5.1	(36)	
	5% WB	MCWBR	4.6	5.0	7.0	7.0	6.4	6.6	5.6	6.3	6.2	5.4	4.7	4.3	(37)	
		MCWBR	5.3	6.5	10.1	12.3	11.9	12.1	12.2	12.4	11.0	8.1	5.9	4.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.330	0.353	0.383	0.418	0.423	0.421	0.430	0.422	0.395	0.367	0.344	0.326	(40)	
		taud	2.378	2.344	2.266	2.201	2.229	2.250	2.228	2.276	2.342	2.403	2.384	2.352	(41)	
	Ebn at Noon		675	760	803	817	832	836	820	806	785	735	649	610	(42)	
		Edh at Noon	63	84	108	130	132	131	131	119	100	78	62	55	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.73	1.41	2.54	4.09	4.94	5.32	5.04	4.29	3.14	1.85	0.87	0.56	(44)	
	RadStd		0.10	0.27	0.38	0.45	0.43	0.46	0.63	0.39	0.34	0.16	0.11	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (86 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page